

Duy M. Dao

[CITY, COUNTRY] · [YOUR EMAIL] · [PHONE] · [GITHUB OR WEBSITE]

Education

Tsinghua University

Beijing, China

[DEGREE], School of Materials Science and Engineering

[YEAR] – [EXPECTED YEAR]

- Research on defect physics in halide perovskites (CsPbI_3) with first-principles calculations and molecular dynamics.

Ho Chi Minh City University of Technology (VNU-HCM)

Ho Chi Minh City, Vietnam

[DEGREE AND MAJOR]

[YEAR] – [YEAR]

- [GPA, HONOURS OR RANK]
- [THESIS TITLE]

Research Experience

Tsinghua University, [GROUP NAME]

Beijing, China

Graduate Researcher

[MONTH YEAR] – Present

- Model the migration of iodine vacancies and interstitials near the CsPbI_3 surface using slab models and nudged elastic band calculations.
- Relaxed the cubic α and orthorhombic γ phases of CsPbI_3 in Quantum ESPRESSO and analysed the transition through the tolerance factor and soft phonon modes.
- Wrote reusable Python scripts that convert CIF structures into ready-to-run Quantum ESPRESSO inputs.
- Run VASP and Quantum ESPRESSO jobs on a PBS-managed HPC cluster.

[EARLIER GROUP OR LAB, IF ANY]

[CITY, COUNTRY]

[ROLE]

[MONTH YEAR] – [MONTH YEAR]

- [WHAT YOU DID AND WHAT CAME OUT OF IT]

Projects

Statistical Mechanics from Maximum Entropy [YEAR]

- Wrote LaTeX lecture notes that derive the ensembles of statistical mechanics from entropy maximisation.

Reproduction of a Machine-Learning Model-Selection Study [YEAR]

- Rebuilt the pipeline of a published paper on NBA betting models that compares model selection by accuracy and by calibration.

Publications

[AUTHORS. TITLE. JOURNAL, VOLUME, PAGES (YEAR). OR REMOVE THIS SECTION]

Talks and Posters

[TITLE. EVENT, CITY (MONTH YEAR). OR REMOVE THIS SECTION]

Awards and Scholarships

[AWARD OR SCHOLARSHIP, AWARDED BODY] [YEAR]

Skills

Simulation: Quantum ESPRESSO, VASP, LAMMPS; DFT, molecular dynamics, NEB

Programming and tools: Python, ASE, VESTA, OVITO, LaTeX, Linux, PBS

Languages: Vietnamese (native), English ([LEVEL]), [OTHER LANGUAGES AND LEVELS]